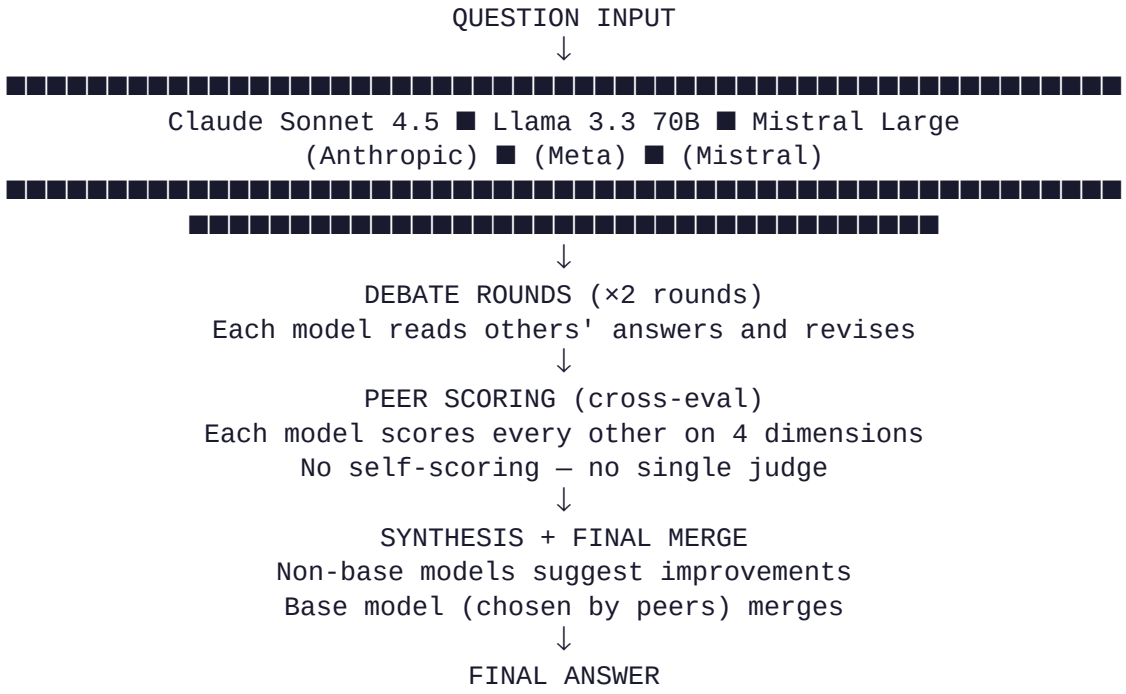
Slide 1: How It Works

Multi-Model Debate Architecture



Key constraint met: No single model decides. All 3 models contribute scores, all non-base models contribute synthesis suggestions. The final answer reflects input from every model.

Built with: Python, LangGraph, AWS Bedrock, 3 models from 3 vendors.

Slide 2: What 'Best' Means

Defining and Measuring Answer Quality

We define answer quality across **four dimensions**, each scored 0–10:

Dimension	Weight	What It Measures
Completeness	25%	Does it address all aspects of the question?
Accuracy	25%	Are claims factually correct and well-supported?
Reasoning	25%	Is the logic clear, structured, and sound?
Nuance	25%	Does it handle edge cases, tradeoffs, complexity?

Why not voting?

The problem explicitly rules out voting: 'Voting works on multiple choice; it does not work on a paragraph.' Our dimensional scoring evaluates the quality of free-text answers without reducing them to binary choices.

How we tested it:

- Independent multi-judge panel (Llama 3 70B + Mistral Large) scores each answer
- Judges are different from debate participants to avoid self-evaluation bias
- Each question: 3 standalone answers + 1 debate answer, all scored blind
- Compare debate score vs. best individual model score per question

Slide 3: The Evidence

Does the Debate Beat Individual Models?

Question set: 8 open-ended questions across multiple categories (strategy, ethics, analysis, creative, policy)

Metric	Value
Questions evaluated	8
Debate wins / ties / losses	3 / 3 / 2
Debate never-worse rate	6/8 (75%)
Avg margin over best individual	+0.19/10
Debate avg score	8.97/10
Claude Sonnet 4.5 avg score	8.77/10
Llama 3.3 70B avg score	8.67/10
Mistral Large avg score	8.47/10

Per-category breakdown:

Category	Win Rate	Avg Margin
strategy	25%	+0.09
ethics	0%	+0.00
analysis	100%	+0.62
creative	0%	-0.12
policy	100%	+0.62

Conclusion: Debate won 3/8, tied 3, lost 2. Avg margin: +0.19/10. Wins (+0.50 to +0.63) were much larger than losses (-0.13). Debate adds most value on complex analytical and policy questions.

Slide 4: Value to a Professional Services Firm

The Problem in Professional Services

Consultants, analysts, and advisors answer complex, open-ended questions daily — strategy recommendations, risk assessments, policy analysis. A single AI model gives one perspective. Clients pay for **thorough, multi-perspective analysis**.

What This System Delivers

- **Higher-quality first drafts:** Debate-synthesized answers are more complete and nuanced than any single model — reduces revision cycles
- **Built-in devil's advocate:** Models challenge each other's reasoning, catching blind spots a single model misses
- **Auditable reasoning:** Full transcript shows how the answer was reached — critical for regulated industries and client trust
- **Vendor diversity:** No single-vendor lock-in; if one model degrades, the system still functions with the remaining models
- **Scalable expertise:** Run 100 debates in parallel — impossible with human panels

Use Cases

- Due diligence analysis (M&A;, vendor evaluation)
- Policy recommendation memos
- Risk assessment with competing viewpoints
- Technical architecture decision records
- Client-facing research where thoroughness matters

Slide 5: Measuring the Return

Baseline

Single-model answer (best available model, e.g., Claude Sonnet 4.5). This is what teams use today — one prompt, one answer. Measure: quality score (same 4-dimension rubric), time-to-deliverable, revision count.

Metrics to Track

Metric	Baseline (Single Model)	Debate System	How to Measure
Answer quality	Avg score on rubric	Avg score on rubric	Blind human eval + automated judges
Revision cycles	Typical: 2-3 rounds	Target: 0-1 rounds	Count before client acceptance
Time to first draft	Seconds	Minutes (3-5x longer)	Wall clock per question
Cost per question	~$0.01-0.05	~$0.15-0.50	Token cost from spend tracker
Coverage of perspectives	1 viewpoint	3+ viewpoints synthesized	Rubric: nuance dimension

Kill Criteria

Kill the project if after 2 weeks of production use:

- Debate win rate < 50% on real client questions (not worth the cost)
- Cost per question exceeds $1.00 without proportional quality gain
- Human reviewers prefer single-model output ≥60% of the time in blind eval
- Latency (>5 min per question) makes it unusable in time-sensitive workflows

Slide 6: Two More Weeks

Current Design Choices

The system uses **TF-IDF cosine similarity** for adaptive early stopping (skips debate rounds when models converge), **quality-weighted peer scoring** (stronger models get more vote weight), **adversarial debate prompts** (challenge before synthesize), **dimension-targeted synthesis** (focuses on the weakest-scoring dimension), and a **self-consistency merge gate** (reverts if the merge degrades quality). A **$5.00 spend ceiling** is enforced per run.

With Two More Weeks

Priority	Improvement	Technique / Reference	Expected Impact
1	Parallel model calls	asyncio.gather() / LangGraph Send()	3-4x faster (latency)
2	Stronger eval judges	Chatbot Arena pairwise comparison (Zheng et al., 2023)	More reliable scoring
3	Error handling + fallbacks	LangChain .with_retry() + .with_fallbacks()	Essential for production
4	Structured output for scoring	JSON mode / with_structured_output()	Zero score parse failures
5	Role-differentiated debate	Critic / steelman / synthesizer roles (Society of Mind, Liang 2023)	+0.2-0.5 on nuance
6	Semantic convergence	Sentence-transformers (MiniLM) or BERTScore (Zhang 2020)	Better early-stop detection
7	Pipeline checkpointing	LangGraph MemorySaver / SqliteSaver	Resume crashed pipelines
8	Human eval study	Blind A/B pairs with consultants	Validate real-world value

Where the Model Was Wrong and I Overrode It

1. Judge self-evaluation bias: Claude initially configured Claude Sonnet as both a debate participant and an evaluation judge — the same model scoring its own output. I overrode this: judges must be independent. The final config uses Llama 3 70B and Mistral Large as judges.

2. Convergence detection: The initial implementation used Jaccard word-overlap similarity, which scored Claude's round-over-round changes at 0.23 — suggesting massive divergence. Cosine similarity on the same text scored 0.73, correctly detecting that the meaning was stable despite rephrasing. I replaced Jaccard with TF-IDF cosine similarity.

3. Consensus bias in prompts: The original debate prompt told models to 'incorporate the best insights from all responses' — which biased toward averaging and diluted strong positions. I rewrote it to require models to challenge each other's weakest points first, only adopting ideas that survive scrutiny.